

Project Report: Cloud-Based QMS & Calibration Automation

END-TO-END QUALITY ENGINEERING ARCHITECTURE CASE STUDY

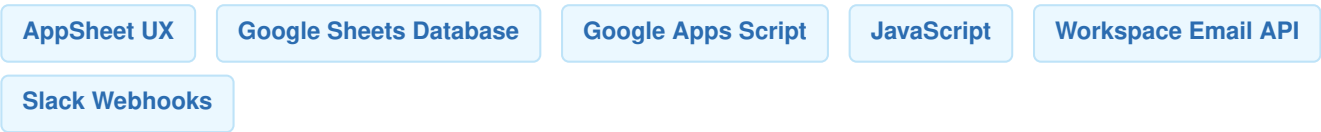
Project Title:	Automated Tool & Calibration Management System	Target Compliance:	ISO 9001 / Traceability Standard
Developer:	Quality / Systems Automation Engineering Portfolio	Primary Stack:	AppSheet, Apps Script, Google Workspace API

1. Executive Summary

In high-precision electronics manufacturing environments, relying on manual, paper-based tracking grids or isolated spreadsheet logs frequently leads to visibility gaps, out-of-compliance risk, and potential production downtime. This engineering project documents the end-to-end design, implementation, and deployment of a mobile-first, cloud-based **Quality Management System (QMS)**.

The finalized ecosystem transitions the facility's asset verification from a human-dependent reactive pipeline into a fail-safe, automated framework. Key structural accomplishments include embedding real-time multi-point electrical tolerance parameter calculators ($\pm 0.05\%$) directly into shop-floor forms, deploying programmatic server-side notification engines, and establishing seamless cloud references to streamline compliance audits.

2. Technical Stack & Architecture



The platform utilizes a structured relational data model combined with cloud automation interfaces to create an instant synchronization loop across shop floor hardware and managing servers.

3. Comprehensive System Walkthrough (A to Z)

Phase A: Relational Database Topography

A multi-table structured schema was engineered within the cloud core database to handle asset lifecycles, maintenance intervals, and technical audit paths safely without record overlap:

- **ExactTools_Inventory_Master:** The core relational authority managing unique hardware Asset IDs, serial details, current deployment location fields, and a dynamic lifecycle state matrix (Active, Expired, In Calibration, or Out of Service).
- **Internal_Cal_Log:** A transactional data table tracking individual measurement items, live device values, user metadata, and evaluated accuracy differentials.
- **Inspection_Records:** A separate historical logging ledger explicitly dedicated to routine mechanical preventatives and physical visual checks.

Phase B: Mobile-First Operator Interfaces

The front-end client exposes a segmented, desktop-and-mobile optimized dashboard with 4 specialized operational portals:

- **Ext Cal Cert Upload:** Configured for primary reference standards (e.g., Siglent DMMs, Fluke Calibrators) managed by ISO-certified testing facilities. Operators load certificate fields and index the updated calibration baseline dates.
- **Exact Tools Inventory Count:** A live, searchable asset deck utilizing real-time filtering queries. Operators perform rapid asset ID, status, and precise workshop-location inquiries.
- **Tool PM Checklist:** Houses the monthly visual and mechanical preventative safety routines, ensuring physical hardware parameters remain within operating specs.
- **Internal Verification Check:** The high-precision mathematical data logging portal where live physical values are validated against structural references during active floor inspections.

Phase C: Technical Standardization & Verification Protocols

Standard operating procedures (SOPs) were converted into programmatic restrictions within the data logging flows to strictly prevent manual estimation errors:

- **Mechanical Fail-Safes:** Integrated strict checks on bed-of-nails assemblies, specifically target validation for pogo pin return elasticity, visual PCB frame centering, and structural mounting fasteners. This ensures early detection of worn or stripped screws to eliminate microscopic plate drift.
- **Precision Tolerancing:** Outlined automated evaluation gates for advanced functional fixtures (such as the EM3 and Flex PCBA Fixtures). The system handles multi-rail verification steps across precision sensor-simulating resistor components down to the milliohm.

Component Ref	Nominal Value (Target)	Tolerance Threshold (\$ ± 0.05%)	Allowable Guarded Windows
R13 (Low-Ohm Reference)	\$0.10 Ω	± 0.00005 Ω	\$0.09995 - 0.10005 Ω (Requires 4-Wire Mode)
R14	\$27.005 kΩ	± 0.0135 kΩ	\$26.9915 - 27.0185 kΩ
R15	\$9.997 kΩ	± 0.0050 kΩ	\$9.9920 - 10.0020 kΩ
R16	\$2.2007 kΩ	± 0.0011 kΩ	\$2.1996 - 2.2018 kΩ
R19	\$4.9913 kΩ	± 0.0025 kΩ	\$4.9888 - 4.9938 kΩ
R22	\$274 kΩ	± 0.1370 kΩ	\$273.8630 - 274.1370 kΩ

Phase D: Server Automation & Notification Logic

The engine incorporates autonomous cloud triggers to handle risk management and maintain operational guardrails:

- **Instant Fail-Safe Lockout:** If a technician logs an out-of-tolerance metric and submits a FAIL condition, an AppSheet automation rule intercepts the submission. The asset state is instantly flipped to a locked status, and an emergency alert dispatch is routed directly to the engineering lead with diagnostic detail data.
- **Predictive Chronological Scans:** A server-side **Google Apps Script** evaluates database timestamps every night at midnight. It automatically flags metrics nearing expiration deadlines, executing proactive notification updates (a 30-day lookahead window for external laboratory calibrations and a 7-day alert trigger for internal mechanical PM deadlines).
- **Cloud URL Record Optimization:** Rearchitected document access parameters by moving from restricted device file buffers to direct Google Drive reference URL endpoints. Technicians open verification sheets instantly in-browser, bypassing download cycles and assuring full version accuracy.

4. Quantifiable Quality Metrics & Portfolio Results

- **Continuous Compliance Traceability:** Established an unbroken audit link connecting standard equipment measurements directly to accredited regulatory reference points.
- **Proactive Downtime Prevention:** Transitioned the facility from reactive, failure-based maintenance to a predictive framework, maintaining high floor operational up-time.

- **Audit Readiness Acceleration:** Programmed streamlined historical compilation views, reducing compliance inspection preparation times down from multiple hours to a single button click.